

Suggested Teaching Guidelines for
AI Compute Platforms, Applications & Trends
PGCP-AI August 2026

Duration: 180 Hours: 60 Theory (T) + 60 Lab (L)+ 60 Self-Learning (SL)

Objective: The objective of this module is to make students aware with the latest AI tools and Techniques used by IT Industry

Prerequisites: Knowledge of fundamentals SQL queries, programming and basic mathematics. ML & statistical knowledge

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment– 20%

List of Course material

Text Book: No specific courseware for the module, faculty may share some coursematerials

Reference Book:

1. Cloud Computing: Master The Concepts, Architecture and Applications With Real-World Examples And Case Studies by Mehul Mahrishi Kamal Kant Hiran, Ruchi Doshi, Dr. Fagbola Temitayo
2. Kubernetes: Up and Running - Dive into the Future of Infrastructure, Second Edition [Paperback] Brendan Burns; Kelsey Hightower and Joe Beda by Brendan Burns (Author), Kelsey Hightower (Author), Joe Beda (Author)

Note:

- **Applications/case studies are advised to be covered under Lab Hours.**
- **Faculty is advised to give appropriate assignment for sufficient hands-on.**
- **Faculty is advised to give at-least one end-to-end problem statement where students can execute their learning in a proper manner.**

I.PySpark (28 Hours of Theory + 22 Hours of Lab + 24 Hours of Self-Learning)

Sessions 1 & 2: (6T + 4L + 3SL)

Theory:

- Apache Spark APIs for large-scale data processing
- Overview, Linking with Spark,
- Initializing Spark
- Spark session & spark context
- Resilient Distributed Datasets (RDDs), External Datasets, RDD Operations,
- Passing Functions to Spark, Working with Key-Value Pairs, Shuffle operations,
- RDD Persistence, Removing Data, Shared Variables, Deploying to a Cluster
- Spark Data Frames

Lab Assignments:

- Run the provided Hadoop Streaming program using python
- Install Spark

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Sessions 3 & 4: (6T + 4L + 3SL)

Theory:

- EDA using PySpark
- ETL jobs using sparks

Lab Assignments:

- Explore Data using PySpark

Session 5: (4T + 4L +3SL)

Theory:

- Introduction to Kafka
- Working with Kafka using Spark
- Spark streaming

Lab Assignments:

- Install Kafka
- Execute the spark streaming with Kafka

Session 6: (4T + 2L + 3SL)

Theory:

- Integration of setting Spark and Kafka
- Setting up Kafka Producer and Consumer
- Kafka Connect API

Lab Assignments:

- Connect the Kafka with API

Session 7: (2T + 2L + 3SL)

Theory:

- Machine Learning using Spark's MLlib

Lab Assignments:

- Explore MLlib

Session 8: (2T + 2L + 3SL)

Theory:

- Deep Learning using Spark
- Introduction to FastAPI
- FastAPI for AI/ML Model Deployment
- High-Performance API Serving using FastAPI

Lab Assignments:

- Explore Deep Learning
- Create a FastAPI application
- Expose a trained Spark MLlib model through a FastAPI endpoint

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Session 9: (2T + 2L + 3SL)**Theory:**

- Spark SQL

Lab Assignments:

- Run the Spark SQL programs using step –runs for each and every code line
- Analyze the election data using spark and provide analysis

Session 10: (2T + 2L + 3SL)**Theory:**

- Connecting DB's with Spark
- Working with DB's

Lab Assignments:

- Connecting DB's with Spark
- Accessing and manipulating the DB's
- Create a complex workflow using bash operator, a simple workflow using python

II.DevOps for AI/ML (32 Hours of Theory and 38 Hours of Lab +36 Hours of Self-Learning)**Sessions 11 & 12: Git / GitHub (4T + 4L + 5SL)****Theory:**

- Introduction to Version control systems
- Creating GitHub repository
- Using Git – Introduction to Git commands

Lab Assignment:

- Create a GitHub repository from the command line

Sessions 13 & 14: Introduction to containers (4T + 6L + 5SL)**Theory:**

- Introduction to DevOps
- Introduction to Containers
- Advantages of using container-based applications
- Installing Docker and using basic Docker commands
- Build your own container-based application image
- Networking in Docker
- Managing containers – Logs / Resources

Lab Assignments:

- Create an httpd container running on port 8000. Name the container as web5. Do not map any directory. Create an "index.html" file & copy this file inside the container and check if the website displays your webpage. Again, modify the "index.html" file and copy it inside the container, and check if the website is updated or not.
- Test any program on-base machine. Then create an image using Dockerfile to execute this exe program. Give image name as "Name: v1".

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Sessions 15 & 16: Introduction to Kubernetes (4T + 6L + 5SL)**Theory:**

- Need for Kubernetes
- Introduction to Kubernetes cluster – Basic terms - Management node, Worker Nodes, Pods, Deployment, and Service Types etc.
- Working with Kubernetes Cluster – Creating deployment, Exposing Deployment as a service, managing your applications. Rolling application updates etc.

Lab assignment:

- Create a Kubernetes cluster, deploy and explore any app.

Sessions 17 & 18: CI/CD with Jenkins (4T + 6L + 5SL)**Theory:**

- Introduction to CI/CD
- Using Jenkins to build a CI/CD pipeline.

Lab Assignments:

- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the NGINX image.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the httpd image.

Sessions 19 & 20: (4T + 4L +4SL)**Theory:**

- Cloud Computing Basics, Understanding Cloud Vendors (AWS/Azure/GCP), Definition, Characteristics, Components

Lab Assignments:

- Study about cloud and other similar configuration
- Explore available solutions
- Cloud Architecture

Sessions 21 & 22: (4T + 4L +4SL)**Theory:**

- Cloud provider, SAAS, PAAS, IAAS and other Organizational scenarios of clouds, Administering & Monitoring cloud services, benefits, and limitations

Lab Assignments:

- Provide a solution on cloud as SAAS using available systems.

Sessions 23 & 24: (4T + 4L +4SL)**Theory:**

- Deploy application over cloud. Comparison among SAAS, PAAS, IAAS, Cloud Products and Solutions, Cloud Pricing, Compute Products and Services, Elastic Cloud Compute, Dashboard.

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Lab Assignments:

- Deploy any application over the cloud platform.

Sessions 25 & 26: (4T + 4L +4SL)

Theory:

- Exploring cloud services for AI/ML

Lab Assignments:

- Explore the different cloud computing platforms for AI/ML.